Secure Hybrid VLC-RF Systems With Light Energy Harvesting

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***Abstract*— In this paper, a hybrid visible light communication-radio frequency (RF) system, including a legitimate receiver (R) and an eavesdropper (E) is considered. R can harvest energy from the light emitted by light emitting diodes (LEDs), which is used for the information transmission between R and the RF receiver which is close to the LED. It is assumed that E tries to eavesdrop the information delivered from R to the RF receiver and R is with finite energy storage. Considering the randomness of the locations of R and E, the statistical characteristics of the received signal-to-noise ratio at the RF receiver and E are characterized; then, we derive the analytical expressions for exact and asymptotic secrecy outage probability by using the stochastic geometry method. Finally, simulations are carried out to verify our proposed analytical models.**

***Index Terms*— Light energy harvesting, Rician fading, secrecy outage probability, stochastic geometry, visible light communication.**

I. Introduction

OVER the past few years, visible light communication (VLC) has been presented as a promising technology to satisfy the growing demand of high data rate mobile services and seamless coverage, especially in indoor scenarios, which was confronted with by traditional radio frequency (RF) communication systems. VLC can support high speed data transmissions, while only using a vast unregulated and free spectrum and requiring the existing illumination infrastructure, i.e., light emitting diodes (LED), to realize wireless data delivery [1].

In order to exploit the advantages of LED and to implement VLC in practical, many research efforts have been dedicated to studying VLC systems in some aspects, e.g., channel modeling, hardware design, modulation scheme, performance modeling and transmission scheme design.

Because of the inherent broadcast nature of VLC channels, the information transmission may probably be eavesdropped by unauthorized terminals in the coverage of the LED, while light signals are normally transmitted without optical fibers or any sort of wave-guiding. Then, physical-layer (PHY) security has been emerging as a promising way to protect the information-theoretic security from eavesdropping [2]–[8].

There are some works proposed to study and improve the secrecy performance of VLC systems [9]–[13]. In [9], beamforming was used to obtain a closed-form secrecy rate expression for the multiple-input single-output (MISO) wiretap channel. Transmit beamformers were designed to maximize the achievable secrecy rate while subjecting to the amplitude constraints imposed by the limited dynamic range of LEDs [10]. Mukherjee [11] analyzed the upper and lower bounds for the single-input single-output VLC secret-key capacities, as well as an MISO VLC secret-key transmission strategy. Instead of maximizing VLC secrecy rate, Shen *et al.* [12] studied the secrecy-oriented transmitter optimization from a quality of service aspect for a VLC MISO system with multiple eavesdroppers. Though the randomness of the positions of the terminals are considered while analyzing the secrecy performance of VLC systems in [13], only the eavesdropping over the VLC downlink are studied. Therefore, it is clear to see that all aforementioned works are limited to VLC systems and can not be adopted to hybrid VLC-RF systems.

We can also find that most of the works related to VLC only focus on the transmission over the downlinks from the LED to the terminals. In practical scenarios, the terminals may need to transmit its data to the LED, e.g., the data collected by the sensors. However, it is normally impractical for the sensors with limited size and constrained battery capacity to work as a power-hungry light source [14]. Furthermore, VLC over the uplinks may produce glare, which is uncomfortable to and undesirable for human. To overcome these problems, RF delivery, e.g., wireless fidelity (Wi-Fi), has considered for the uplinks as a suitable alternative for VLC.

So far, researchers have studied and designed hybrid VLC-RF systems to exploit the advantages of VLC and RF [15]–[20], e.g., the capability of supporting high-speed data transmission by VLC and achieving seamless coverage by RF. An optimal bit-and-power allocation algorithm was proposed for VLC-orthogonal frequency-division multiplexing systems in [15]. Wang *et al.* [16] studied heterogeneous

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VLC-WiFi systems, in which VLC aims at supporting high data rate transmission and RF for reliable connectivity. Shao *et al.* [17] designed and analyzed a hybrid WiFi-VLC system, which utilizes VLC and WiFi as the downlink and the uplink, respectively. Load balancing in a hybrid light fidelity/Wi-Fi network was studied in [18] while considering user mobility and handover signalling overheads. In [19], a hybrid VLC/RF downlink was studied, which can realize full RF-coverage by making use of WiFi and the abundant spectral resources offered by VLC. Bao *et al.* [20] proposed a hybrid VLC/orthogonal frequency-division multiplexing access network model. Therefore, one can easily observe that the PHY security of hybrid VLC-RF systems has not been fully exploited yet.

In indoor scenarios, there exists some energy-constrained terminals, e.g., sensors for monitoring the temperature, humidity and indoor air quality, etc. Then, it is crucial to improve and extend the lifetime of these terminals which are with limited energy budget. So far, there are several schemes presented for wireless sensors to harvest energy from various kinds of energy sources in the physical world, including solar energy, mechanical vibration, temperature gradient, wind energy, water flow, and magnetic energy. Then, the light emitted by the LED lamp can be used for energy harvesting [21]. Ramos *et al.* [22] adopted visible light both for wake-up communication and energy harvesting purposes. The medium access control protocol for indoor VLC with energy harvesting was explored in [23]. Light energy harvesting has been considered at the relay in a dual-hop heterogeneous VLC/RF communication system in [24]. However, none of these works has studied the mathematical model for the energy harvesting in VLC systems.

Motivated by these observations, in this work, using stochastic geometry theory, we study the secrecy outage performance of a hybrid VLC-RF system consisting of a legitimate receiver and an eavesdropper, while considering the light energy harvesting over the downlink and the randomness of the locations of the legitimate receiver and the eavesdropper.

In our previous work [13], the secrecy performance of the VLC downlink was studied while considering the randomness of the positions of the terminals. In this work, the secrecy outage performance of the RF uplink in a hybrid VLC-RF system is investigated while considering the randomness of the positions of the legitimate receiver and the eavesdropper and light energy harvesting.

The main contributions of our work are summarized as follows:

- We characterize the probability density function (PDF) and cumulative distribution function (CDF) of the signal-to-noise-ratio (SNR) over the VLC downlink and RF uplink;
- Light energy harvesting model has been proposed for VLC systems;
- The analytical expressions for the exact and asymptotic secrecy outage probability (SOP) have been derived.

The paper is structured as follows. In Section II, the considered system model is presented. The analysis on the exact secrecy performance of the information transmissions over the RF uplink is presented in Section III. In Section IV, the asymptotic SOP is proposed. The proposed analytical models are verified by Monte-Carlo simulations in Section V. Finally, the paper is concluded in Section VI.

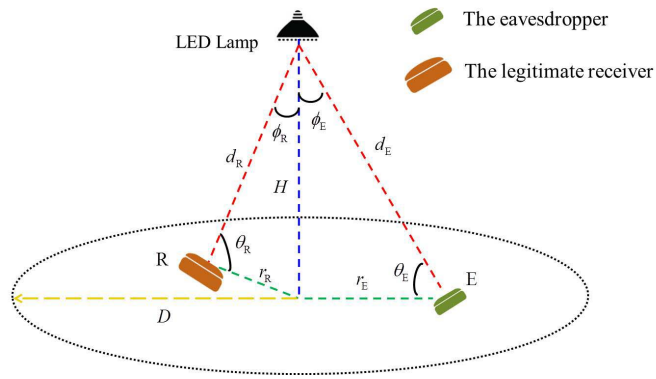

Fig. 1. System model.

II. System Model

In this work, we consider a hybrid VLC-RF system, as shown in Fig. 1, where a group of L ($L \geq 1$) LED lamps are placed in the center of the ceiling to insure the illumination and to deliver data to a legitimate receiver (R), while there is an eavesdropper (E) locating in the same area and trying to eavesdrop the information transmission between the LED lamps and R. Moreover, R and E are with photodiode (PD) for the purposes of data communication and light energy harvesting.

For tractability purpose, in this work we also treat the ground floor of the room as a disc,[1] $\mathcal{S}$, with radius, D, where the projection of the LED lamps is located at the center of the disc. Without of loss generality, we assume that R and E are uniformly distributed in the disc.

The information delivery between R and the LED can be divided into two stages:

- One of these LED lights transmits requirement messages over the downlink to R, while the other LED lights just play the role of illumination. Then, in this stage R can harvest energy from the light emitted by all LED lights;
- R sends its information back to the RF receiver at the LED by using the traditional RF communication, while E trying to overhear the information delivery.

In this work, we assume that the uplink between R and the RF receiver suffers Rician fading[2] and the channel sate

[1] In practice, the illumination projection of a light on the floor is a disc, where the light is placed at the center of the ceiling and the floor area is usually larger than the coverage area of the light. Then, the used assumption can be justified because it is reasonable for the eavesdropper to stay in the coverage of the LED lamp and close to R to maximize its eavesdropping capability. Moreover, there also exists a case that the floor area of the room is larger than the coverage of a LED lamp. Then, the difference between the coverage of the LED lamp and the floor area of the room will lead to approximation errors, which will be numerically discussed in Section V. In this work, we only focus on the case that the floor area of the room is similar to the coverage of the LED lamp, and we leave the case that the floor area of the room is larger than the coverage of the LED lamp as a future research direction.

[2] Rician fading is a specialized model for stochastic fading when there are multipath fading signals including a line of sight (LoS) signal, which is much stronger than the others.

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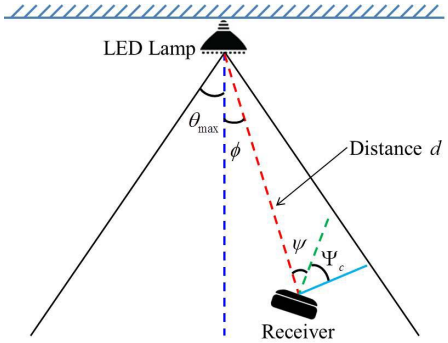

Fig. 2. LOS channel model.

information of the uplink is available at R. Furthermore, to promise the success of the information delivery over the uplink, R carries on the information transmission to the RF receiver only when the received SNR at the RF receiver is larger than a predefined threshold; otherwise, R will abandon information transmitting and keep silence.

Then, in this work we focus on the secure information delivery over the uplink. We also denote the time duration for the downlink between the ith ($1 \le i \le L$) lamp and R as T_{dn}, and the one for uplink from R to the RF receiver as T_{up}. There is η ($0 \le \eta \le 1$) portion of T_{dn} used for the data transmission from the ith ($1 \le i \le L$) lamp to R.

A. Channel Model for VLC

There is a LoS path from the LED to each node, and also a diffuse path via the reflections from surfaces within the room, as shown by Fig. 2. In this work, we only consider LoS propagation and omit reflections from surrounding surfaces for simplification. The channel gain from the transmitter to the receiver is given by [25]

$$h = \begin{cases} \dfrac{(m+1)A}{2\pi d^2}\cos^m(\phi)\, T_s(\psi)\, g(\psi)\cos\psi, & 0 \le \psi \le \Psi_c \\ 0, & \psi > \Psi_c, \end{cases} \tag{1}$$

where d is the distance between the LED lamp and the receiver, A is the area of the photodetector, ϕ is the transmitter viewing angle (also referred as irradiation angle), ψ is the angle of incidence with respect to the receiver axis, Ψ_c is the concentrator field of view, $T_s(\psi)$ denotes the gain of the optical filter adopted at the receiver, m denotes the order of Lambertian emission, which is related to $\phi_{1/2}$ (the semi-angle of the LED), the transmitter semi-angle (at half power), by $m = -\ln 2/\ln\left(\cos\left(\phi_{1/2}\right)\right)$, $g(\psi)$ denotes the gain of the non-imaging concentrator, which is given by [25, eq. (8)]. In this work, it is assumed that the average optical transmit power, P_t, and the area of the photodetector at all LED lamps are same.

B. Light Energy Harvesting Model

The received optical power from the ith ($1 \le i \le L$) LED lamp at node j ($j \in \{\mathrm{R}, \mathrm{E}\}$) can be given as

$$P_{ij,R} = P_t \frac{(m_i+1)A}{2\pi {d_j}^2}\cos^{m_i}\left(\phi_{ij}\right) T_s\left(\psi_{ij}\right) g\left(\psi_{ij}\right)\cos\left(\psi_{ij}\right), \tag{2}$$

where h_{ij}, m_i, ϕ_{ij}, ψ_{ij} and d_j are the parameters defined in Eq. (1) for the jth receiver. It is obvious that $P_{ij,R}$ is not only related to the distance between the ith lamp and node j ($j \in \{\mathrm{R}, \mathrm{E}\}$), but also related to both the radiation pattern of the ith ($1 \le i \le L$) lamp and the incidence angle of node j ($j \in \{\mathrm{R}, \mathrm{E}\}$).

As suggested by [26, eq. (11)], the RF power from the ith ($1 \le i \le L$) lamp at node j ($j \in \{\mathrm{R}, \mathrm{E}\}$) can be written as

$$P_{ij,RF} = C_{j,RF} P_t^2 G_t^2\left(\phi_{ij}\right) G_r^2\left(\psi_{ij}\right)/d_j^4, \tag{3}$$

where $C_{j,RF}$ is the RF power constant at the jth receiver, which is related to the optical-to-electrical conversion efficiency and optical power constant given the following relation $\left(P_{j,RF}\right)^{\frac{1}{2}} \propto i_{PDj} \propto P_{rj}$ (i_{PDj} is the current of the PD at mode j), $G_t\left(\phi_{ij}\right)$ and $G_r\left(\psi_{ij}\right)$ are the transmitting and receiving gains related to the radiation angle ϕ_{ij} and incidence angle ψ_{ij}, respectively, which can be given as $G_t\left(\phi_{ij}\right) = \cos^{m_{i,t}}\left(\phi_{ij}\right)$ and $G_r\left(\psi_{ij}\right) = \cos^{m_{ij,r}}\left(\psi_{ij}\right)$, where $m_{i,t}$ and $m_{ij,r}$ are related to $\phi_{i,1/2}$ and $\psi_{ij,1/2}$, which are the ith lamp and the jth receiver semi-angles at half power, and are given by $m_{i,t} = -\ln 2/\ln\left(\cos\left(\phi_{i,1/2}\right)\right)$ and $m_{ij,r} = -\ln 2/\ln\left(\cos\left(\psi_{ij,1/2}\right)\right)$, respectively.

In the following, we assume the incidence line is perpendicular to the PD axis at each receiver to maximize the light receiving.[3] Then, it deduces $\psi_{ij} = 0$ for $1 \le i \le L$. It is also assumed that $\phi_{ij} = \phi_j$, $m_{i,t} = m_t$ ($1 \le i \le L$) to simplify the analysis, as all LED lamps are close to each other. Therefore, Eq. (3) can be rewritten as

$$P_{ij,RF} = C_{j,RF} P_t^2 \Delta\cos^{2m_t}\left(\phi_j\right)/d_j^4, \tag{4}$$

where $\Delta = \cos^{2m_{ij,r}}\left(\psi_\Delta\right)$ ($0 < \Delta \le 1$) is the orientation error of the PD, which is due to the estimated error of ψ_{ij} (ψ_Δ, $0 \le \psi_\Delta < \psi_{ij}$), when the positional information of R is imperfect.

The harvested energy at R can be given as

$$\begin{aligned} E_{\mathrm{R}} &= P_{i\mathrm{R},RF} T_{dn}\rho(1-\eta) + \sum_{k\in\Pi, k\ne i} P_{k\mathrm{R},RF} T_{dn}\rho \\ &= \frac{P_t^2 \Delta T_{dn}\rho}{d_{\mathrm{R}}^4}\Bigg[C_{\mathrm{R},RF}(1-\eta)\cos^{2m_t}\left(\phi_{\mathrm{R}}\right) \\ &\quad + \sum_{k\in\Pi, k\ne i} C_{\mathrm{R},RF}\cos^{2m_{k,t}}\left(\phi_{k\mathrm{R}}\right)\Bigg] \\ &= \frac{P_t^2 \Delta T_{dn}\rho C_{\mathrm{R},RF}(L-\eta)}{d_{\mathrm{R}}^4}\cos^{2m_t}\left(\phi_{\mathrm{R}}\right), \end{aligned} \tag{5}$$

where ρ ($0 < \rho < 1$) is the energy harvesting efficiency at R, $N_0 = N_{0,j}$ and $\Pi = \{1, 2, \cdots, L\}$.

C. Signal Model for the Uplink

In this work, we assume that R is with a finite energy storage size B.

[3] In practical scenarios, after each realization of the deployment of the system, R can adjust its PD's direction $\theta_{\mathrm{R}} = \arccos\left(\frac{r_R}{\sqrt{H^2+r_R^2}}\right)$ degree from the horizontal direction to promise the incidence line is perpendicular to the PD axis.

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Then, the average transmitting power at R over the uplink can be expressed as

$$P_{\mathrm{R}}=\begin{cases}\dfrac{E_{\mathrm{R}}}{T_{up}}=\dfrac{P_t^2\Delta T_{dn}\rho C_{\mathrm{R},RF}\left(L-\eta\right)}{d_{\mathrm{R}}^4T_{up}}\cos^{2m_t}\left(\phi_{\mathrm{R}}\right) & E_{\mathrm{R}}\leq B;\\ \dfrac{B}{T_{up}} & E_{\mathrm{R}}>B,\end{cases} \tag{6}$$

where

$$\Pr\left\{E_{\mathrm{R}}\leq B\right\}=\Pr\left\{\frac{P_t^2\Delta T_{dn}\rho C_{\mathrm{R},RF}\left(L-\eta\right)}{d_{\mathrm{R}}^4}\cos^{2m_t}\left(\phi_{\mathrm{R}}\right)\leq B\right\}.$$

(1) When $E_{\mathrm{R}}\leq B$

Observing Fig. 1, one has $\cos\left(\phi_j\right)=H/\sqrt{H^2+r_j^2}$ and $d_j=\sqrt{H^2+r_j^2}$, then $P_{ij,RF}$ can be rewritten as $P_{ij,RF}=C_{j,RF}P_t^2\Delta H^{2m_t}\left(H^2+r_j^2\right)^{-(m_t+2)}$.

The received SNR at node j ($j\in\{\mathrm{R},\mathrm{E}\}$) can be written as $\gamma_{ij}=\frac{P_{ij,RF}}{N_{0,j}}=\lambda_j\left(H^2+r_j^2\right)^{-(m_t+2)}$, where $N_{0,j}$ is the power of the noise at the jth receiver, $\lambda_j=C_{j,RF}P_t^2\Delta H^{2m_t}/N_{0,j}$. In the following, we let $\gamma_j=\gamma_{ij}$ for simplification.

Then, P_{R} and $\Pr\left\{E_{\mathrm{R}}\leq B\right\}$ can be rewritten as

$$\begin{aligned}P_{\mathrm{R}}&=\frac{E_{\mathrm{R}}}{T_{up}}\\&=\frac{P_t^2\Delta T_{dn}\rho H^{2m_t}C_{\mathrm{R},RF}\left(L-\eta\right)}{T_{up}}\cdot\frac{1}{\left(H^2+r_{\mathrm{R}}^2\right)^{(m_t+2)}}\end{aligned} \tag{7}$$

and

$$\begin{aligned}\Pr\left\{E_{\mathrm{R}}\leq B\right\}&=\Pr\left\{\frac{P_t^2\Delta T_{dn}\rho H^{2m_t}C_{\mathrm{R},RF}\left(L-\eta\right)}{\left(H^2+r_{\mathrm{R}}^2\right)^{(m_t+2)}}\leq B\right\}\\&=\Pr\left\{\frac{\upsilon}{\left(H^2+r_{\mathrm{R}}^2\right)^{(m_t+2)}}\leq B\right\},\end{aligned} \tag{8}$$

respectively, where $\upsilon=P_t^2\Delta T_{dn}\rho H^{2m_t}C_{\mathrm{R},RF}\left(L-\eta\right)$.

Then, the received signal at the RF receiver from the legitimate receiver over the uplink can be given as

$$y=\sqrt{\frac{P_{\mathrm{R}}}{{d_{\mathrm{R}}}^n}}h_{\mathrm{R}}x_{\mathrm{R}}+z, \tag{9}$$

where h_{R} is the channel gain from R to the RF receiver, x_{R} is the information data transmitted by R, n is the path loss exponent and z is the complex Gaussian noise at the RF receiver with average power N_0.

The received signal at E from R can be given as

$$y_{\mathrm{RE}}=\sqrt{\frac{P_{\mathrm{R}}}{{d_{\mathrm{RE}}}^n}}h_{\mathrm{RE}}x_{\mathrm{R}}+z_{\mathrm{E}}, \tag{10}$$

where h_{RE} is the channel gain from R to E, d_{RE} is the distance between R and E, which can be written as $d_{\mathrm{RE}}=\sqrt{r_{\mathrm{R}}^2+r_{\mathrm{E}}^2-2r_{\mathrm{R}}r_{\mathrm{E}}\cos\left(\varphi_{\mathrm{R}}-\varphi_{\mathrm{E}}\right)}$ (where φ_{R} and φ_{E} are the angular coordinates for the locations of R and E in polar coordinate, respectively), z_{E} is the complex Gaussian noise at E with average power N_0.

Then, the received SNR at the RF receiver and E can be written as

$$\begin{aligned}\gamma_0&=\frac{P_{\mathrm{R}}|h_{\mathrm{R}}|^2}{N_0{d_{\mathrm{R}}}^n}\\&=\frac{P_t^2\Delta T_{dn}\rho C_{\mathrm{R},RF}\left(L-\eta\right)}{N_0T_{up}}\cdot\frac{|h_{\mathrm{R}}|^2}{d_{\mathrm{R}}^{n+4}}\cdot\cos^{2m_t}\left(\phi_{\mathrm{R}}\right)\\&=\tau|h_{\mathrm{R}}|^2\left(H^2+r_{\mathrm{R}}^2\right)^{-(m_t+n/2+2)}\end{aligned} \tag{11}$$

and

$$\begin{aligned}\gamma_{\mathrm{RE}}&=\frac{P_{\mathrm{R}}|h_{\mathrm{RE}}|^2}{N_0{d_{\mathrm{RE}}}^n}\\&=\frac{P_t^2\Delta T_{dn}\rho C_{\mathrm{R},RF}\left(L-\eta\right)}{N_0T_{up}}\cdot\frac{|h_{\mathrm{RE}}|^2}{d_{\mathrm{R}}^4d_{\mathrm{RE}}^n}\cdot\cos^{2m_t}\left(\phi_{\mathrm{R}}\right)\\&=\tau|h_{\mathrm{RE}}|^2\left(H^2+r_{\mathrm{R}}^2\right)^{-m_t-2}\\&\quad\cdot\left(r_{\mathrm{R}}^2+r_{\mathrm{E}}^2-2r_{\mathrm{R}}r_{\mathrm{E}}\cos\left(\varphi_{\mathrm{R}}-\varphi_{\mathrm{E}}\right)\right)^{-\frac{n}{2}},\end{aligned} \tag{12}$$

respectively, where $\tau=\frac{P_t^2\Delta T_{dn}\rho H^{2m_t}C_{\mathrm{R},RF}(L-\eta)}{N_0T_{up}}$.

(2) When $E_{\mathrm{R}}>B$

In this case, $P_{\mathrm{R}}=B/T_{up}$. Then, the received SNR at the RF receiver and E can be written as

$$\begin{aligned}\gamma_0&=\frac{P_{\mathrm{R}}|h_{\mathrm{R}}|^2}{N_0{d_{\mathrm{R}}}^n}=\frac{B}{T_{up}N_0}\cdot\frac{|h_{\mathrm{R}}|^2}{d_{\mathrm{R}}^n}\\&=\lambda|h_{\mathrm{R}}|^2\left(H^2+r_{\mathrm{R}}^2\right)^{-n/2}\end{aligned} \tag{13}$$

and

$$\begin{aligned}\gamma_{\mathrm{RE}}&=\frac{P_{\mathrm{R}}|h_{\mathrm{RE}}|^2}{N_0{d_{\mathrm{RE}}}^n}=\frac{B}{T_{up}N_0}\cdot\frac{|h_{\mathrm{RE}}|^2}{d_{\mathrm{RE}}^n}\\&=\lambda|h_{\mathrm{RE}}|^2\left(r_{\mathrm{R}}^2+r_{\mathrm{E}}^2-2r_{\mathrm{R}}r_{\mathrm{E}}\cos\left(\varphi_{\mathrm{R}}-\varphi_{\mathrm{E}}\right)\right)^{-n/2},\end{aligned} \tag{14}$$

respectively, where $\lambda=B/(T_{up}N_0)$.

To promise the information transmission between R and the RF receiver at the LED, the received SNR at the RF receiver, γ_0, should be larger than a predefined threshold, γ_{th}; otherwise, outage event will occur as R will stop transmitting information to the RF receiver.

Using Eqs. (11) and (13), we can obtain $\Pr\left\{\gamma_0\geq\gamma_{\mathrm{th}}\right\}$ as

$$\Pr\left\{\gamma_0\geq\gamma_{\mathrm{th}}\right\}=\begin{cases}\Pr\left\{\tau|h_{\mathrm{R}}|^2\left(H^2+r_{\mathrm{R}}^2\right)^{-(m_t+n/2+2)}\geq\gamma_{\mathrm{th}}\right\} & E_{\mathrm{R}}\leq B;\\ \Pr\left\{\lambda|h_{\mathrm{R}}|^2\left(H^2+r_{\mathrm{R}}^2\right)^{-n/2}\geq\gamma_{\mathrm{th}}\right\} & E_{\mathrm{R}}>B.\end{cases} \tag{15}$$

III. Secrecy Outage Analysis

The instantaneous secrecy capacity can be presented as

$$C_s\left(\gamma_0,\gamma_{\mathrm{RE}}\right)=\max\left\{\log_2\left(1+\gamma_0\right)-\log_2\left(1+\gamma_{\mathrm{RE}}\right),0\right\}. \tag{16}$$

Also, in this work the SOP is defined as the probability that the secrecy capacity is equal to zero.

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When $\gamma_0 \geq \gamma_{\text{th}}$, the information delivery will be carried on between R and the RF receiver. Then, in this case the SOP can be written as

$$\begin{aligned} P_o &= \Pr\left\{\log_2(1+\gamma_0) - \log_2(1+\gamma_{\text{RE}}) \leq 0, \gamma_0 \geq \gamma_{\text{th}}\right\} \\ &= \Pr\left\{\gamma_0 \leq \gamma_{\text{RE}}, \gamma_0 \geq \gamma_{\text{th}}\right\}. \end{aligned} \tag{17}$$

Therefore, it is easy to obtain the SOP for $E \leq B$ and $E > B$ as

$$\begin{aligned} P_o &= \Pr\left\{\tau|h_{\text{R}}|^2\left(H^2+r_{\text{R}}^2\right)^{-(m_t+n/2+2)}\right. \\ &\leq \tau|h_{\text{RE}}|^2\left(H^2+r_{\text{R}}^2\right)^{-m_t-2} \\ &\quad \left.\cdot\left(r_{\text{R}}^2+r_{\text{E}}^2-2r_{\text{R}}r_{\text{E}}\cos(\varphi_{\text{R}}-\varphi_{\text{E}})\right)^{-\frac{n}{2}}, \gamma_0 \geq \gamma_{\text{th}}\right\} \\ &= \Pr\left\{\frac{|h_{\text{R}}|^2}{|h_{\text{RE}}|^2} \leq \left(\frac{H^2+r_{\text{R}}^2}{r_{\text{R}}^2+r_{\text{E}}^2-2r_{\text{R}}r_{\text{E}}\cos(\varphi_{\text{R}}-\varphi_{\text{E}})}\right)^{\frac{n}{2}}\right\} \\ &\quad \cdot\Pr\left\{\gamma_0 \geq \gamma_{\text{th}}\right\} \end{aligned} \tag{18}$$

and

$$\begin{aligned} P_o &= \Pr\left\{\lambda|h_{\text{R}}|^2\left(H^2+r_{\text{R}}^2\right)^{-\frac{n}{2}} \leq \lambda|h_{\text{RE}}|^2\right. \\ &\quad \left.\cdot\left(r_{\text{R}}^2+r_{\text{E}}^2-2r_{\text{R}}r_{\text{E}}\cos(\varphi_{\text{R}}-\varphi_{\text{E}})\right)^{-\frac{n}{2}}, \gamma_0 \geq \gamma_{\text{th}}\right\} \\ &= \Pr\left\{\frac{|h_{\text{R}}|^2}{|h_{\text{RE}}|^2} \leq \left(\frac{H^2+r_{\text{R}}^2}{r_{\text{R}}^2+r_{\text{E}}^2-2r_{\text{R}}r_{\text{E}}\cos(\varphi_{\text{R}}-\varphi_{\text{E}})}\right)^{\frac{n}{2}}\right\} \\ &\quad \cdot\Pr\left\{\gamma_0 \geq \gamma_{\text{th}}\right\}, \end{aligned} \tag{19}$$

respectively.

By observing Eqs. (18) and (19), one can see that the SOP for the case that $\gamma_0 \geq \gamma_{\text{th}}$ is irrelevant to the relationship between the harvested energy and the energy storage size.

Therefore, while considering the outage events occurring in both cases: $\gamma_0 < \gamma_{\text{th}}$ and $\gamma_0 \geq \gamma_{\text{th}}$, we can write the SOP for the considered system as

$$\begin{aligned} P_{out} &= \Pr\{E_{\text{R}} \leq B\}\left[W_1 + (1-W_1)(1-T)\right] \\ &\quad + \Pr\{E_{\text{R}} > B\}\left[W_2 + (1-W_2)(1-T)\right], \end{aligned} \tag{20}$$

where

$$W_1 = \Pr\left\{\tau|h_{\text{R}}|^2\left(H^2+r_{\text{R}}^2\right)^{-\frac{2m_t+n+4}{2}} < \gamma_{\text{th}}\right\},$$

$$W_2 = \Pr\left\{\lambda|h_{\text{R}}|^2\left(H^2+r_{\text{R}}^2\right)^{-\frac{n}{2}} < \gamma_{\text{th}}\right\},$$

$$T = \Pr\left\{\frac{|h_{\text{R}}|^2}{|h_{\text{RE}}|^2} > \left(\frac{H^2+r_{\text{R}}^2}{r_{\text{R}}^2+r_{\text{E}}^2-2r_{\text{R}}r_{\text{E}}\cos(\varphi_{\text{R}}-\varphi_{\text{E}})}\right)^{\frac{n}{2}}\right\}.$$

In this work, we assume that the channels among R, the RF receiver and E follow Rician distribution. Then, the PDF and CDF of $|h_l|^2$ ($l \in \{\text{R}, \text{RE}\}$) can be given as [27]

$$\begin{aligned} f_{|h_l|^2}(x) &= \frac{(1+K)}{\Omega}\cdot\exp\left(-K-\frac{x(1+K)}{\Omega}\right) \\ &\quad \cdot I_0\left(2\sqrt{\frac{K(1+K)x}{\Omega}}\right), x \geq 0 \end{aligned} \tag{21}$$

and

$$F_{|h_l|^2}(x) = 1 - Q_1\left(\sqrt{2K}, \sqrt{\frac{2(1+K)x}{\Omega}}\right), \tag{22}$$

respectively, where Ω is the variance of the signal, K is Rician factor which corresponds to the ratio of the power of the LOS (specular) component to the average power of the scattered component, $Q_1(a,b) = \int_b^\infty x\cdot\exp\left(-\frac{x^2+a^2}{2}\right)I_0(ax)\,dx$ is Marcum-Q function, $I_0(\cdot)$ is modified Bessel function of order 0.

The PDF of r_j ($j \in \{\text{R}, \text{E}\}$) can be obtained as $f(x) = \frac{2x}{D^2}$, $0 \leq x \leq D$. Further we can achieve the PDF and CDF of $\omega = \left(H^2+r_j^2\right)^{-\alpha}$ ($j \in \{\text{R}, \text{E}\}$, α is a positive constant) as

$$f_\omega(x) = a_0 x^{b_0} \tag{23}$$

and

$$F_\omega(x) = \frac{a_0}{c_0}x^{c_0} + t, \tag{24}$$

respectively, where $\upsilon_{\min} = \left(H^2+D^2\right)^{-\alpha} \leq x \leq \upsilon_{\max} = H^{-2\alpha}$, $a_0 = -c_0/D^2$, $b_0 = c_0 - 1$, $c_0 = -\alpha^{-1}$ and $t = H^2/D^2 + 1$.

Thus, it is easy to have

$$\begin{aligned} \Pr\{E_{\text{R}} \leq B\} &= \Pr\left\{\frac{\upsilon}{\sqrt[2m_t+4]{H^2+r_{\text{R}}^2}} \leq B\right\} \\ &= \Pr\left\{\left(H^2+r_{\text{R}}^2\right)^{-m_t-2} \leq \frac{B}{\upsilon}\right\} \\ &= \begin{cases} 0 & \frac{B}{\upsilon} < \upsilon_{\min} \\ 1 + \frac{H^2}{D^2} - \frac{1}{D^2}\left(\frac{\upsilon}{B}\right)^{\frac{1}{m_t+2}} & \upsilon_{\min} \leq \frac{B}{\upsilon} \leq \upsilon_{\max} \\ 1 & \frac{B}{\upsilon} > \upsilon_{\max} \end{cases} \end{aligned} \tag{25}$$

$$\begin{aligned} \Pr\{E_{\text{R}} > B\} &= \Pr\left\{\frac{\upsilon}{\sqrt[2m_t+4]{H^2+r_{\text{R}}^2}} > B\right\} \\ &= \begin{cases} 1 & \frac{B}{\upsilon} < \upsilon_{\min} \\ \frac{1}{D^2}\left(\frac{\upsilon}{B}\right)^{\frac{1}{m_t+2}} - \frac{H^2}{D^2} & \upsilon_{\min} \leq \frac{B}{\upsilon} \leq \upsilon_{\max} \\ 0 & \frac{B}{\upsilon} > \upsilon_{\max} \end{cases} \end{aligned} \tag{26}$$

$$\begin{aligned} W_1 &= 1 - Q_1\left(\sqrt{2K}, \sqrt{\frac{2\gamma_{\text{th}}(1+K)}{\Omega_R\tau}}\right. \\ &\quad \left.\cdot\left(H^2+r_{\text{R}}^2\right)^{\frac{2m_t+n+4}{4}}\right) \end{aligned} \tag{27}$$

$$\begin{aligned} W_2 &= 1 - Q_1\left(\sqrt{2K}, \sqrt{\frac{2\gamma_{\text{th}}(1+K)}{\lambda\Omega_E}}\right. \\ &\quad \left.\cdot\left(H^2+r_{\text{R}}^2\right)^{\frac{n}{4}}\right), \end{aligned} \tag{28}$$

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where Ω_D and Ω_E are the variances of the signal over the channel between R and the RF receiver, and the one between R and E, respectively.

In order to calculate Eqs. (27) and (28), we will give a useful theorem as follows:

Theorem 1: Let $g(x) = Q_1\left(p, qx^k\right)$ $\left(H^{2k} \leq x \leq \left(H^2+D^2\right)^k\right)$, where p, q and k are constant, the PDF of x is $f(x) = ax^{-b}$, where $a = -1/(kD^2)$ and $b = k^{-1} - 1$. Then, we obtain $I(p,q,k) = \int_{H^{2k}}^{\left(H^2+D^2\right)^k} Q_1\left(p, qx^k\right) f(x)\, dx$ as

$$I(p,q,k) = \frac{a}{uk}\left(\frac{1}{e^v q^u}\right)^{-\frac{b-1}{uk}}\left[\Gamma\left(-\frac{b-1}{uk}, e^v q^u H^{2uk^2}\right) - \Gamma\left(-\frac{b-1}{uk}, e^v q^u\left(H^2+D^2\right)^{uk^2}\right)\right], \quad (29)$$

where $u = 2.174 - 0.592p + 0.593p^2 - 0.092p^3 + 0.005p^4$, $v = -0.840 + 0.327p - 0.740p^2 + 0.083p^3 - 0.004p^4$ and $\Gamma(\alpha, x) = \int_x^\infty t^{\alpha-1}\exp(-t)\, dt$ is the upper incomplete gamma function.

Proof: Please refer to Appendix A. ■

Then, considering the randomness of the locations of R and E, and using **Theorem 1**, we can derive Eqs. (27) and (28) as

$$W_1 = 1 - I\left(\sqrt{2K}, \sqrt{\frac{2\gamma_{\text{th}}(1+K)}{\Omega_R \tau}}, \frac{2m_t + n + 4}{4}\right) \quad (30)$$

and

$$W_2 = 1 - I\left(\sqrt{2K}, \sqrt{\frac{2\gamma_{\text{th}}(1+K)}{\lambda\Omega_R}}, \frac{n}{4}\right), \quad (31)$$

respectively.

In the following, we will calculate the item, $T = \Pr\left\{\frac{|h_{\text{R}}|^2}{|h_{\text{RE}}|^2} > \left(\frac{H^2+r_{\text{R}}^2}{r_{\text{R}}^2+r_{\text{E}}^2-2r_{\text{R}}r_{\text{E}}\cos(\varphi_{\text{R}}-\varphi_{\text{E}})}\right)^{\frac{n}{2}}\right\}$ in Eq. (20).

Lemma 1: Considering the the Rician fading over the channels among R, the RF receiver and E, T can be obtained as

$$T = \sum_{k=0}^{\infty}(-1)^k \exp(-2K)\frac{L_k^{(0)}(K)}{\Gamma(k+2)}\Gamma(k+2) \cdot \left(\frac{H^2+r_{\text{R}}^2}{r_{\text{R}}^2+r_{\text{E}}^2-2r_{\text{R}}r_{\text{E}}\cos(\varphi_{\text{R}}-\varphi_{\text{E}})}\right)^{-\frac{n}{2}(k+1)} \cdot {}_1F_1(k+2;1;K), \quad (32)$$

where $L_k^{(0)}(x) = \sum_{l=0}^{k}\frac{k!}{(k-l)!l!}(-x)^l$ is the generalized Laguerre polynomial with degree k and order 0, given by [28, eq. (22.3.9)].

Proof: See Appendix B. ■

Next, we will calculate T while taking the randomness of R and E into account.

For frequencies between 800 MHz and 1.9 GHz, COST 231 model reports [29], [30]: $n = 2, 4$ and 6 for large open rooms, densely furnished rooms and the rooms with cluttered and obstructed paths, respectively. For tractability, in this work we only consider cases that the path loss factor is an even number. Then, $\frac{n}{2}(k+1)$ is a positive integer. In the following, we denote $m = \frac{n}{2}(k+1)$.

As R and E are uniformly distributed in the disc, $\mathcal{S}$, then we have φ_{R} and φ_{E} are uniformly distributed in the interval $(0, 2\pi]$, and the PDF of φ_{R} and φ_{E}, $f_{\varphi_j}(x) = \frac{1}{2\pi}$ $(j \in \{\text{R}, \text{E}\})$. Recalling that the PDF of r_j $(j \in \{\text{R}, \text{E}\})$ can be obtained as $f(x) = \frac{2x}{D^2}$, $0 \leq x \leq D$, we can rewrite T as

$$\begin{aligned} T &= \sum_{k=0}^{\infty}(-1)^k e^{-2K}\frac{L_k^{(0)}(K)}{\Gamma(k+2)}\Gamma(k+2)\,{}_1F_1(k+2;1;K) \\ &\quad \cdot \frac{1}{\pi^2 D^4}\int_0^{2\pi}\int_0^{2\pi}\int_0^{D}\int_0^{D}\left(\frac{r_{\text{R}}^2+r_{\text{E}}^2-2r_{\text{R}}r_{\text{E}}\cos(\varphi_{\text{R}}-\varphi_{\text{E}})}{H^2+r_{\text{R}}^2}\right)^m \\ &\quad \cdot r_{\text{R}}r_{\text{E}}dr_{\text{R}}dr_{\text{E}}d\varphi_{\text{R}}d\varphi_{\text{E}} \\ &= \sum_{k=0}^{\infty}(-1)^k e^{-2K}\frac{L_k^{(0)}(K)}{\Gamma(k+2)}\Gamma(k+2)\,{}_1F_1(k+2;1;K) \\ &\quad \cdot \sum_{p=0}^{m}\binom{m}{p}\frac{(-1)^{m-p}}{\pi^2 D^4}\int_0^{D}\int_0^{D}\left(\frac{r_{\text{R}}^2+r_{\text{E}}^2}{H^2+r_{\text{R}}^2}\right)^p\left(\frac{2r_{\text{R}}r_{\text{E}}}{H^2+r_{\text{R}}^2}\right)^{m-p} \\ &\quad \cdot r_{\text{R}}r_{\text{E}}\int_0^{2\pi}\int_0^{2\pi}\cos^{m-p}(\varphi_{\text{R}}-\varphi_{\text{E}})\, d\varphi_{\text{R}}d\varphi_{\text{E}}dr_{\text{R}}dr_{\text{E}}. \end{aligned} \quad (33)$$

Lemma 2: Using the sum-difference formula of cosine function and [31, eq. (2.5.12.44)], the inner double-integral in last equation, $T_{in1} = \int_0^{2\pi}\int_0^{2\pi}\cos^{m-p}(\varphi_{\text{R}}-\varphi_{\text{E}})\, d\varphi_{\text{R}}d\varphi_{\text{E}}$, can be derived as

$$T_{in1} = \sum_{q=0}^{m-p}\binom{m-p}{q}(-1)^{m-p-q}[g(p,q)]^2, \quad (34)$$

where $g(p,q) = \frac{[1+(-1)^{m-p-q}][1+(-1)^q]}{2}\text{B}\left(\frac{m-p-q+1}{2}, \frac{q+1}{2}\right)$, $\text{B}(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}dt$ is the beta function.

Proof: See Appendix C. ■

Lemma 3: Considering the randomness of r_R and r_E, the double-integral of r_R and r_E in Eq. (33), $T_{in2} = \int_0^D\int_0^D\left(\frac{r_{\text{R}}^2+r_{\text{E}}^2}{H^2+r_{\text{R}}^2}\right)^p\left(\frac{2r_{\text{R}}r_{\text{E}}}{H^2+r_{\text{R}}^2}\right)^{m-p}dr_{\text{R}}dr_{\text{E}}$, can be derived as

$$T_{in2} = \sum_{l=0}^{p}\binom{p}{l}\frac{2^{m-p}}{m+p-2l+2}\cdot\frac{1}{2l+m-p+2}\frac{D^{2m+4}}{H^{2m}} \cdot {}_2F_1\left(m, \frac{2l+m-p}{2}+1; \frac{2l+m-p}{2}+2; -\frac{D^2}{H^2}\right), \quad (35)$$

where ${}_2F_1$ is Gauss hypergeometric function, as defined by [28, eq. (9.100)].

Proof: See Appendix D. ■

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Theorem 2: The analytical expression for $T = \Pr\left\{\frac{|h_{\mathrm{R}}|^2}{|h_{\mathrm{RE}}|^2} > \left(\frac{H^2+r_{\mathrm{R}}^2}{r_{\mathrm{R}}^2+r_{\mathrm{E}}^2-2r_{\mathrm{R}}r_{\mathrm{E}}\cos(\varphi_{\mathrm{R}}-\varphi_{\mathrm{E}})}\right)^{\frac{n}{2}}\right\}$ can be expressed as

$$\begin{aligned} T = &\frac{1}{\pi^2}\sum_{k=0}^{\infty}\exp(-2K)\frac{L_k^{(0)}(K)}{\Gamma(k+2)}\Gamma(k+2) \\ &\cdot {}_1F_1(k+2;1;K)\sum_{p=0}^{m}\binom{m}{p}\sum_{q=0}^{m-p}\binom{m-p}{q}\frac{D^{2m}}{H^{2m}} \\ &\cdot(-1)^{m+k-p}2^{m-p}[g(p,q)]^2 \\ &\cdot\sum_{l=0}^{p}\binom{p}{l}\frac{1}{m+p-2l+2}\cdot\frac{1}{2l+m-p+2} \\ &\cdot {}_2F_1\left(m,\frac{2l+m-p}{2}+1;\frac{2l+m-p}{2}+2;-\frac{D^2}{H^2}\right). \end{aligned} \tag{36}$$

Proof: The analytical expression for T can be easily obtained by by using **Lemma 2** and **Lemma 3** into **Lemma 1**.

Then, the proof is completed. ■

Finally, the SOP can be achieved by substituting Eqs. (25)-(26), (30)-(31) and (36) into Eq. (20).

IV. Asymptotic Analysis

In this section, we present the asymptotic analysis for the SOP while $H \to 0$. [4]

Lemma 4: When $H \to 0$, the probability that the harvested energy at R is not larger than B can be rewritten as

$$\Pr\left\{E_{\mathrm{R},asy} \le B\right\} = \begin{cases} 0 & \upsilon_{asy} > BD^4 \\ 1-\frac{1}{D^2}\sqrt{\frac{\upsilon_{asy}}{B}} & \upsilon_{asy} \le BD^4. \end{cases} \tag{37}$$

Proof: When $H \to 0$, $\cos(\phi_{\mathrm{R}}) \to 1$. Then, the harvested energy at R in this case can be written as

$$\begin{aligned} E_{\mathrm{R},asy} &= P_{i\mathrm{R},RF}T_{dn}\rho(1-\eta) + \sum_{k\in\Pi,k\neq i}P_{k\mathrm{R},RF}T_{dn}\rho \\ &= \frac{P_t^2T_{dn}\rho}{d_{\mathrm{R}}^4}\left[C_{\mathrm{R},RF}(1-\eta)+\sum_{k\in\Pi,k\neq i}C_{\mathrm{R},RF}\right] \\ &= \frac{\upsilon_{asy}}{d_{\mathrm{R}}^4}, \end{aligned} \tag{38}$$

where $\upsilon_{asy} = P_t^2T_{dn}\rho C_{\mathrm{R},RF}(L-\eta)$.

As the CDF of r_{R} can be obtained as $F(x) = \begin{cases} 1 & x > D \\ \frac{x^2}{D^2} & 0 \le x \le D \end{cases}$. Therefore, under this case the probability that the harvested energy at R is not larger than B can be rewritten as

$$\begin{aligned} \Pr\left\{E_{\mathrm{R},asy} \le B\right\} &= \Pr\left\{\frac{\upsilon_{asy}}{d_{\mathrm{R}}^4} \le B\right\} \\ &= \begin{cases} 0 & \upsilon_{asy} > BD^4 \\ 1-\frac{1}{D^2}\sqrt{\frac{\upsilon_{asy}}{B}} & \upsilon_{asy} \le BD^4. \end{cases} \end{aligned} \tag{39}$$

Then, the proof is completed. ■

Therefore, $\Pr\left\{E_{\mathrm{R},asy} > B\right\}$ can be obtained by using $\Pr\left\{E_{\mathrm{R},asy} > B\right\} = 1 - \Pr\left\{E_{\mathrm{R},asy} \le B\right\}$ and **Lemma 4** as

$$\Pr\left\{E_{\mathrm{R},asy} > B\right\} = \begin{cases} 1 & \upsilon_{asy} > BD^4 \\ \frac{1}{D^2}\sqrt{\frac{\upsilon_{asy}}{B}} & \upsilon_{asy} \le BD^4. \end{cases} \tag{40}$$

Lemma 5: Let $g(x) = Q_1\left(p, qx^k\right)$ $(H^{2k} \le x \le \left(H^2+D^2\right)^k)$, where p, q and k are constant, the PDF of x is $f(x) = ax^{-b}$, where $a = -1/(kD^2)$ and $b = k^{-1} - 1$. Then, when $H \to 0$, we can obtain the asymptotic result of $I(p,q,k) = \int_{H^{2k}}^{(H^2+D^2)^k} Q_1\left(p, qx^k\right) f(x)\,dx$ as

$$\begin{aligned} I'(p,q,k) = &\frac{a}{uk}\left(\frac{1}{e^{\upsilon}q^u}\right)^{-\frac{b-1}{uk}}\left[\Gamma\left(-\frac{b-1}{uk}\right)\right. \\ &\left.-\Gamma\left(-\frac{b-1}{uk}, e^{\upsilon}q^uD^{2uk^2}\right)\right]. \end{aligned} \tag{41}$$

Proof: $I'(p,q,k)$ can be easily achieved by setting $H = 0$ in Eq. (29).

Then, the proof is completed. ■

Therefore, using **Lemma 5**, we can derive the asymptotic results of Eqs. (27)-(28) as

$$W_{1,asy} = 1 - I'\left(\sqrt{2K}, \sqrt{\frac{2\gamma_{\mathrm{th}}(1+K)}{\Omega\tau}}, \frac{2m_t+n+4}{4}\right) \tag{42}$$

and

$$W_{2,asy} = 1 - I'\left(\sqrt{2K}, \sqrt{\frac{2\gamma_{\mathrm{th}}(1+K)}{\lambda\Omega}}, \frac{n}{4}\right), \tag{43}$$

respectively.

Lemma 6: When $H \to 0$, we can obtain the asymptotic result of $T = \Pr\left\{\frac{|h_{\mathrm{R}}|^2}{|h_{\mathrm{RE}}|^2} > \left(\frac{H^2+r_{\mathrm{R}}^2}{r_{\mathrm{R}}^2+r_{\mathrm{E}}^2-2r_{\mathrm{R}}r_{\mathrm{E}}\cos(\varphi_{\mathrm{R}}-\varphi_{\mathrm{E}})}\right)^{\frac{n}{2}}\right\}$ as

$$\begin{aligned} T_{asy} = &\frac{1}{\pi^2}\sum_{k=0}^{\infty}\exp(-2K)\frac{L_k^{(0)}(K)}{\Gamma(k+2)}\Gamma(k+2){}_1F_1(k+2) \\ &\cdot\sum_{p=0}^{m}\binom{m}{p}\sum_{q=0}^{m-p}\binom{m-p}{q}(-1)^{m+k-p}[g(p,q)]^2 \\ &\cdot\sum_{l=0}^{p}\binom{p}{l}\frac{D^{m+p-2l+1+\frac{2l+m-p+2}{2m}}}{m+p-2l+2} \\ &\cdot\beta\left(\frac{2l+m-p+2}{2m}\right). \end{aligned} \tag{44}$$

[4]Normally, asymptotic analysis is carried on to obtain the performance limitation of the target systems while the transmitted SNR approaching to infinity. In this work, it is more reasonable and practical to present the derivation of the asymptotic SOP by considering $H \to 0$, as the height of the LED, H, is the main factor on the received SNR at the terminal and in practical the transmit power at the LED lights is strictly constrained and cannot approach to infinity.

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Proof: When $H \to 0$, T_{in2} can be derived as

$$T_{in2,asy} = 2^{m-p} \sum_{l=0}^{p} \binom{p}{l} \frac{D^{m+p-2l+2}}{m+p-2l+2} \int_0^D \frac{r_{\mathrm{R}}^{2l+m-p+1}}{1+r_{\mathrm{R}}^{2m}} dr_{\mathrm{R}}$$
$$= 2^{m-p} \sum_{l=0}^{p} \binom{p}{l} \frac{D^{m+p-2l+1+\frac{2l+m-p+2}{2m}}}{m+p-2l+2} \cdot \beta\left(\frac{2l+m-p+2}{2m}\right), \quad (45)$$

where $\beta(x) = \int_0^1 \frac{t^{x-1}}{1+t} dt$, as shown by [28, eq. (8.371.1)].

Therefore, the asymptotic T can be obtained by substituting Eqs. (35) and (45) into Eq. (32).

Then, the proof is completed. ■

So the asymptotic SOP can be obtained by substituting Eqs. (37), (40), (42)-(43) and (44) into Eq. (20).

In the following, we will present the asymptotic SOP for a special case that the channels among R, the RF receiver and E suffer Rayleigh fading, namely, $K \to 0$. Under this case, it is easy to obtain

$$W_{1,ray} = 1 - I'\left(0, \sqrt{\frac{2\gamma_{\mathrm{th}}}{\Omega\tau}}, \frac{2m_t+n+4}{4}\right) \quad (46)$$

$$W_{2,ray} = 1 - I'\left(0, \sqrt{\frac{2\gamma_{\mathrm{th}}}{\lambda\Omega}}, \frac{n}{4}\right) \quad (47)$$

and

$$T_{ray} = \frac{1}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{\Gamma(k+2)} \Gamma(k+2)\, {}_1F_1(k+2) \cdot \sum_{p=0}^{m} \binom{m}{p} \sum_{q=0}^{m-p} \binom{m-p}{q} (-1)^{m+k-p} [g(p,q)]^2 \cdot \sum_{l=0}^{p} \binom{p}{l} \frac{D^{m+p-2l+1+\frac{2l+m-p+2}{2m}}}{m+p-2l+2} \cdot \beta\left(\frac{2l+m-p+2}{2m}\right). \quad (48)$$

Then, the asymptotic SOP for the case of Rayleigh fading can be obtained by substituting Eqs. (37), (40), (46)-(47) and (48) into Eq. (20).

V. Numerical Results

In this section, we run Monte Carlo simulation to validate our proposed expressions for SOP. Unless otherwise explicitly specified, the main parameters are set as $K = 1$, $D = 2$ m, $L = 8$, $\rho = 0.8$, $\eta = 0.1$, $B = 500$ mAh $\times$ 3.0 V, $n = 2$, $T_{dn} = 10^7$ s, $T_R = T_{up}/T_{dn} = 0.01$, $\phi_{1/2} = 70°$, $H = 2.5$ m, $\gamma_{th} = 1$ dB, $N_{0,j} = -98.82$ dBm, $P_t = 6$ W. In this section, we only focus on the case without any positional information error for the adjustment of orienting the PD and let $\Delta = 1$. The truncation factors of the summation in Eqs. (36) and (44) are set as 6. During each case, we run 10^5 trials for the Monte Carlo simulations.

As observed in Fig. 3, the SOP with a small Ω_E outperforms the one with a large Ω_E, as a large Ω_E represents a better channel condition. One can also see from Fig. 3 that simulation results can match well with the analytical ones in case of high Ω_E.

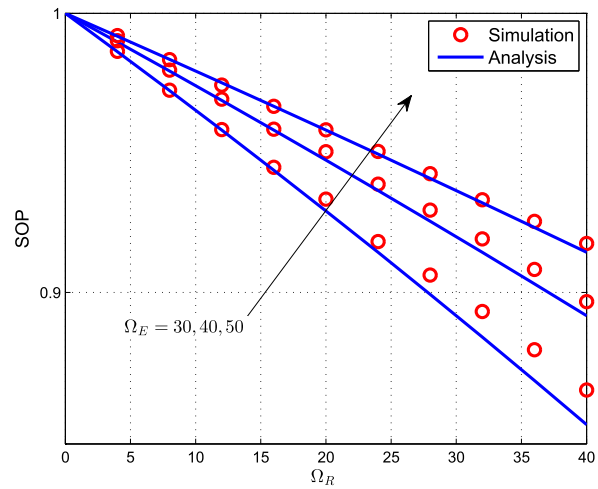

Fig. 3. SOP vs. Ω_R for various Ω_E.

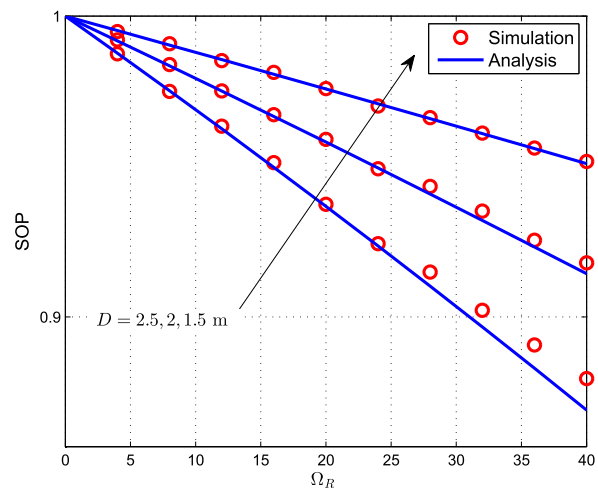

Fig. 4. SOP vs. Ω_R for various D.

As shown in Fig. 4, one can see that the SOP with a small D outperforms the one with a large D. Because a large D means a large distribution area for R and E, leading to an increased probability that the distance between R and E gets larger.

Clearly, as presented in Fig. 5, the SOP with a small H outperforms the one with a large H. It can be explained from the following two aspects: 1) A large H results in a large pathloss over the VLC downlink, which will cause the decreased harvested energy at the receiver; 2) The received signal at the RF receiver over the uplink will degrade while H increases, due to the increased pathloss. However, on the other hand, the received signal over the eavesdropping link between R and E will not be affected by H.

Furthermore, we can find from Figs. 3-5 that simulation results can match well with the analytical ones in low Ω_R region, while the mismatching between simulation and analysis results will enlarge in high Ω_R region. The main reason is the accuracy of the adopted expansion of Marcum Q-function, Eq. (53). However, it is obvious that the derived analysis expression for SOP works well in the case that the condition

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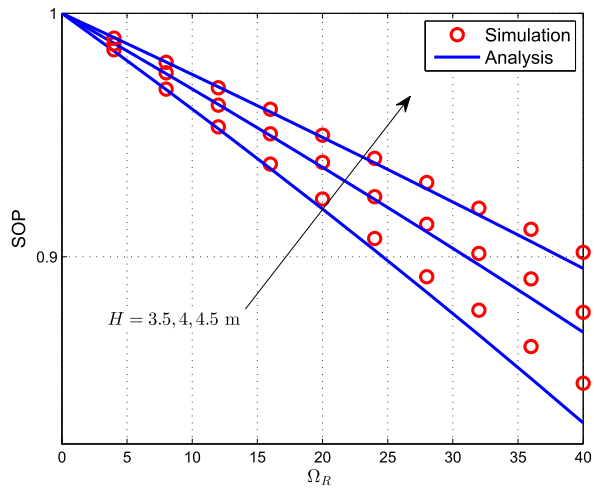

Fig. 5. SOP vs. Ω_R for various H.

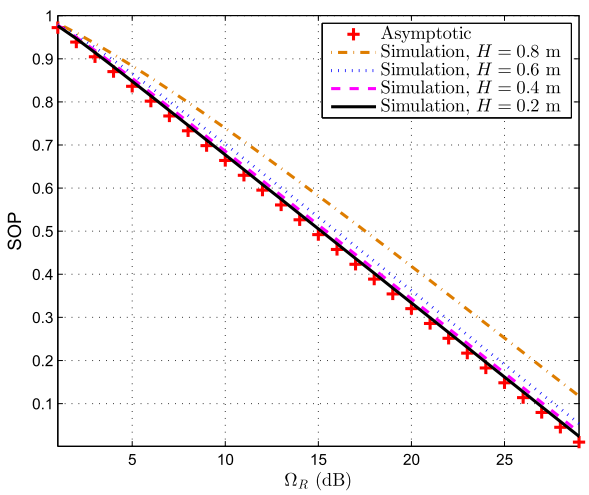

Fig. 6. Asymptotic SOP vs. Ω_R for various H.

of the eavesdropping channel is similar to/outperforms the one of the uplink between R and the RF receiver. This case represents the most practical indoor scenarios. Because in the space-limited indoor scenarios E may always be close to R without any obstacle to achieve the best eavesdropping, then the pathloss of the eavesdropping channel will not be larger than the one of uplink between R and the RF receiver due to the height of the LED light, as indicated in Fig. 1.

Fig. 6 presents the lines for the asymptotic SOP vs. Ω_R for various H, while $\Omega_E = 30$ dB. One can see that the simulation results can approach the asymptotic one closer as H decreases.

Furthermore, as observed in Figs. 3-6, the secrecy performance of the considered system is poor. Because the legitimate receiver and the eavesdropper all locate in a same given room with a limited size, the information delivery over the RF uplink can be easily overheard by the eavesdropper. To improve the secrecy performance, cooperative jamming scheme can be adopted by the RF transceiver when it can work in full-duplex mode [36]. In other words, the RF transceiver can send jamming signals to the eavesdropper by using directional antenna to degrade the received signal-to-interference-plus-noise ratio at the eavesdropper, as the position of the eavesdropper is easily to be available at the RF transceiver.

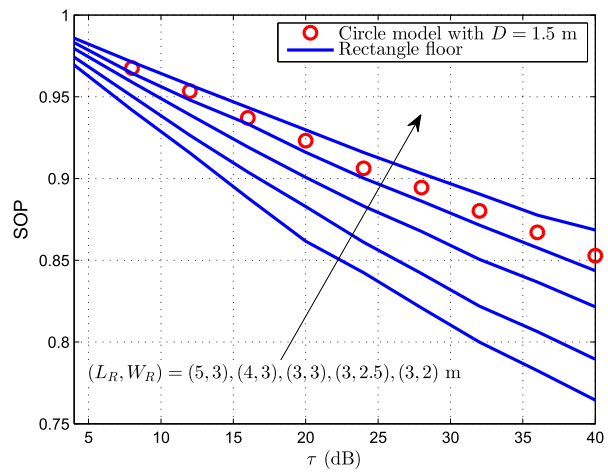

Fig. 7. The approximation accuracy of the adopted circle floor model.

Fig. 7 shows the comparisons among the SOP of the adopted circle floor model and the ones of the rectangle floor with various sizes, where $D = 3$ m, L_R and W_R denote the length and width of the rectangle floor, respectively. One can easily see that the difference among the SOP of the circle floor model and the ones of the rectangle floor increases, when the difference between the circle floor model and the rectangle floor increases. Because, compared to the adopted circle floor model, R and E can locate in a larger area in the case of the rectangle floor with a larger size, then the distance between R and E will increase accordingly, which results in the degraded eavesdropping capacity. Furthermore, it can also be observed that the approximation of our adopted circle floor model is reasonably accurate if the area of the rectangle floor is close to that of the adopted circle floor. Please note that the case that the floor area of the room is larger than the coverage of the LED lamp will be beyond the scope of this paper, but it is an important future research direction.

VI. Conclusion

In this study, we have studied the secrecy outage performance of the uplink in a hybrid VLC-RF system with light energy harvesting, while considering the randomness of the positions of the legitimate receiver and the eavesdropper. Specially, the exact and asymptotic expressions for the secrecy outage probability have been derived in case of finite energy storage size adopted at the legitimate receiver. Numerical results reveal that the radius of the circle has a positive effect on the secrecy outage performance and the height of the LED light has a negative effect on the secrecy out performance. Moreover, our proposed analytical model can work well for the typical case in practical indoor scenarios that the condition of the eavesdropping channel is similar to/outperforms the one of the uplink between R and the RF receiver.

Appendix A
Proof of Theorem 1

As suggested in [32], Marcum Q-function can be presented as

$$Q_1(\alpha, \beta) = \exp\left(-e^{v}\beta^{u}\right), \tag{49}$$

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where $u(\alpha) = 2.174 - 0.592\alpha + 0.593\alpha^2 - 0.092\alpha^3 + 0.005\alpha^4$ and $\upsilon(\alpha) = -0.840 + 0.327\alpha - 0.740\alpha^2 + 0.083\alpha^3 - 0.004\alpha^4$.

Then, we can have $g(x) = Q_1\left(p, qx^k\right) = \exp\left(-e^{\upsilon} q^u x^{uk}\right)$, where $u = 2.174 - 0.592p + 0.593p^2 - 0.092p^3 + 0.005p^4$ and $\upsilon = -0.840 + 0.327p - 0.740p^2 + 0.083p^3 - 0.004p^4$.

By taking the randomness of x into account, we further obtain

$$\begin{aligned} I(p,q,k) &= \int_{H^{2k}}^{\left(H^2+D^2\right)^k} Q_1\left(p, qx^k\right) f(x)\, dx \\ &= a \int_{H^{2k}}^{\left(H^2+D^2\right)^k} \exp\left(-e^{\upsilon} q^u x^{uk}\right) x^{-b} dx. \end{aligned} \tag{50}$$

Let $y = e^{\upsilon} q^u x^{uk}$ ($y_{min} = e^{\upsilon} q^u H^{2uk^2} \le y \le y_{max} = e^{\upsilon} q^u \left(H^2 + D^2\right)^{uk^2}$), then we have $x = \left(\frac{y}{e^{\upsilon} q^u}\right)^{\frac{1}{uk}}$ and $dx = \frac{1}{uk}\left(\frac{1}{e^{\upsilon} q^u}\right)^{\frac{1}{uk}} y^{\frac{1}{uk}-1} dy$. Therefore, one can obtain

$$\begin{aligned} I(p,q,k) &= a \int_{y_{min}}^{y_{max}} \exp(-y) \left(\frac{y}{e^{\upsilon} q^u}\right)^{-\frac{b}{uk}} \frac{1}{uk}\left(\frac{1}{e^{\upsilon} q^u}\right)^{\frac{1}{uk}} \\ &\quad \cdot y^{\frac{1}{uk}-1} dy \\ &= \frac{a}{uk}\left(\frac{1}{e^{\upsilon} q^u}\right)^{-\frac{b-1}{uk}} \int_{y_{min}}^{y_{max}} \exp(-y)\, y^{-\frac{b-1}{uk}-1} dy \\ &= \frac{a}{uk}\left(\frac{1}{e^{\upsilon} q^u}\right)^{-\frac{b-1}{uk}} \left[\Gamma\left(-\frac{b-1}{uk}, e^{\upsilon} q^u H^{2uk^2}\right)\right. \\ &\quad \left. -\Gamma\left(-\frac{b-1}{uk}, e^{\upsilon} q^u \left(H^2 + D^2\right)^{uk^2}\right)\right], \end{aligned} \tag{51}$$

where $\Gamma(\alpha, x) = \int_x^{\infty} t^{\alpha-1} \exp(-t)\, dt$ is the upper incomplete gamma function.

Then, the proof is completed.

APPENDIX B
PROOF OF LEMMA 1

Making use of Eqs. (21) and (22), we can write T as Eq. (52), as presented on the bottom of this page.

As suggested in [33], Marcum Q-function can be presented in terms of generalized Laguerre polynomials as follows

$$Q_1(a,b) = 1 - \sum_{k=0}^{\infty} (-1)^k \exp\left(-\frac{a^2}{2}\right) \frac{L_k^{(0)}\left(\frac{a^2}{2}\right)}{\Gamma(k+2)} \left(\frac{b^2}{2}\right)^{k+1}, \tag{53}$$

where $L_k^{(0)}(x) = \sum_{l=0}^{k} \frac{k!}{(k-l)!l!}(-x)^l$ is the generalized Laguerre polynomial with degree k and order 0, given by [28, eq. (22.3.9)]. A discussion on the truncation error of Eq. (53) can be found in the final table of the precious version of [33] in [34].

Then, using Eq. (53), T can be rewritten as

$$\begin{aligned} T &= \sum_{k=0}^{\infty} (-1)^k \exp(-2K) \frac{L_k^{(0)}(K)}{\Gamma(k+2)} \frac{1+K}{\Omega_R} \left(\frac{1+K}{\Omega_E}\right)^{k+1} \\ &\quad \cdot \left(\frac{H^2 + r_{\mathrm{R}}^2}{r_{\mathrm{R}}^2 + r_{\mathrm{E}}^2 - 2r_{\mathrm{R}} r_{\mathrm{E}} \cos(\varphi_{\mathrm{R}} - \varphi_{\mathrm{E}})}\right)^{-\frac{n}{2}(k+1)} \\ &\quad \cdot \int_0^{\infty} x^{k+1} \cdot \exp\left(-\frac{x(1+K)}{\Omega_R}\right) \cdot I_0\left(2\sqrt{\frac{K(1+K)x}{\Omega_R}}\right) dx, \end{aligned} \tag{54}$$

Let $y = \sqrt{x}$, $dx = 2y dy$. Then, using [35, eq. (2.15.5.4)], the integral in last equation can be written as

$$\begin{aligned} &\int_0^{\infty} x^{k+1} \cdot \exp\left(-\frac{x(1+K)}{\Omega_R}\right) \cdot I_0\left(2\sqrt{\frac{K(1+K)x}{\Omega_R}}\right) dx \\ &= 2\int_0^{\infty} y^{2k+3} \cdot \exp\left(-\frac{(1+K)}{\Omega_R} y^2\right) \\ &\quad \cdot I_0\left(2y\sqrt{\frac{K(1+K)}{\Omega_R}}\right) dy \\ &= \left(\frac{1+K}{\Omega_R}\right)^{-k-2} \Gamma(k+2)\, {}_1F_1(k+2; 1; K), \end{aligned} \tag{55}$$

where $\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} \exp(-t)\, dt$ is the gamma function, ${}_1F_1$ is Gauss hypergeometric function, as defined by [28, eq. (9.100)].

$$\begin{aligned} T &= \Pr\left\{|h_{\mathrm{RE}}|^2 < |h_{\mathrm{R}}|^2 \left(\frac{H^2 + r_{\mathrm{R}}^2}{r_{\mathrm{R}}^2 + r_{\mathrm{E}}^2 - 2r_{\mathrm{R}} r_{\mathrm{E}} \cos(\varphi_{\mathrm{R}} - \varphi_{\mathrm{E}})}\right)^{-\frac{n}{2}}\right\} \\ &= \int_0^{\infty} \left[1 - Q_1\left(\sqrt{2K}, \sqrt{\frac{2(1+K)\left(\frac{H^2 + r_{\mathrm{R}}^2}{r_{\mathrm{R}}^2 + r_{\mathrm{E}}^2 - 2r_{\mathrm{R}} r_{\mathrm{E}} \cos(\varphi_{\mathrm{R}} - \varphi_{\mathrm{E}})}\right)^{-\frac{n}{2}} x}{\Omega_E}}\right)\right] \frac{(1+K)}{\Omega_R} \cdot \exp\left(-K - \frac{x(1+K)}{\Omega_R}\right) \\ &\quad \cdot I_0\left(2\sqrt{\frac{K(1+K)x}{\Omega_R}}\right) dx \end{aligned} \tag{52}$$

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Then, T can be calculated as

$$T = \sum_{k=0}^{\infty} (-1)^k \exp(-2K) \frac{L_k^{(0)}(K)}{\Gamma(k+2)} \Gamma(k+2) \cdot \left(\frac{H^2 + r_{\mathrm{R}}^2}{r_{\mathrm{R}}^2 + r_{\mathrm{E}}^2 - 2r_{\mathrm{R}} r_{\mathrm{E}} \cos(\varphi_{\mathrm{R}} - \varphi_{\mathrm{E}})} \right)^{-\frac{n}{2}(k+1)} \cdot {}_1F_1(k+2; 1; K). \tag{56}$$

Then, the proof is completed.

Appendix C
Proof of Lemma 2

Using the sum-difference formula of cosine function, we can derive the double-integral in Eq. (33), $T_{in1} = \int_0^{2\pi}\int_0^{2\pi} \cos^{m-p}(\varphi_{\mathrm{R}} - \varphi_{\mathrm{E}})\, d\varphi_{\mathrm{R}} d\varphi_{\mathrm{E}}$ as

$$\begin{aligned} T_{in1} &= \int_0^{2\pi}\int_0^{2\pi} \sum_{q=0}^{m-p} \binom{m-p}{q} \cos^q(\varphi_{\mathrm{R}}) \cos^q(\varphi_{\mathrm{E}}) \\ &\quad \cdot \sin^{m-p-q}(\varphi_{\mathrm{R}}) \sin^{m-p-q}(\varphi_{\mathrm{E}})\, d\varphi_{\mathrm{R}} d\varphi_{\mathrm{E}} \\ &= \sum_{q=0}^{m-p} \binom{m-p}{q} (-1)^{m-p-q} \int_0^{2\pi} \cos^q(\varphi_{\mathrm{E}}) \sin^{m-p-q}(\varphi_{\mathrm{E}}) \\ &\quad \cdot \int_0^{2\pi} \cos^q(\varphi_{\mathrm{R}}) \sin^{m-p-q}(\varphi_{\mathrm{R}})\, d\varphi_{\mathrm{R}} d\varphi_{\mathrm{E}}. \end{aligned} \tag{57}$$

To calculate the integral in last equation, we denote a new function, $g(p,q) = \int_0^{2\pi} \cos^q x \sin^{m-p-q} x\, dx$. Then, by using [31, eq. (2.5.12.44)], we can easily have

$$g(p,q) = \frac{\left[1 + (-1)^{m-p-q}\right]\left[1 + (-1)^q\right]}{2} \cdot \mathrm{B}\left(\frac{m-p-q+1}{2}, \frac{q+1}{2}\right), \tag{58}$$

where $\mathrm{B}(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1} dt$ is the beta function.

Therefore, substituting Eq. (58) into Eq. (57), we can rewrite T_{in} as

$$T_{in1} = \sum_{q=0}^{m-p} \binom{m-p}{q} (-1)^{m-p-q} [g(p,q)]^2. \tag{59}$$

Then, the proof is completed.

Appendix D
Proof of Lemma 3

We can calculate the double-integral of r_D and r_E in Eq. (57), $T_{in2} = \int_0^D\int_0^D \left(\frac{r_{\mathrm{R}}^2 + r_{\mathrm{E}}^2}{H^2 + r_{\mathrm{R}}^2}\right)^p \left(\frac{2r_{\mathrm{R}} r_{\mathrm{E}}}{H^2 + r_{\mathrm{R}}^2}\right)^{m-p} r_{\mathrm{R}} r_{\mathrm{E}} dr_{\mathrm{R}} dr_{\mathrm{E}}$, as

$$\begin{aligned} T_{in2} &= 2^{m-p} \sum_{l=0}^{p} \binom{p}{l} \int_0^D\int_0^D \frac{r_{\mathrm{R}}^{2l+m-p+1} r_{\mathrm{E}}^{m+p-2l+1}}{\left(H^2 + r_{\mathrm{R}}^2\right)^m} dr_{\mathrm{R}} dr_{\mathrm{E}} \\ &= 2^{m-p} \sum_{l=0}^{p} \binom{p}{l} \frac{D^{m+p-2l+2}}{m+p-2l+2} \int_0^D \frac{r_{\mathrm{R}}^{2l+m-p+1}}{\left(H^2 + r_{\mathrm{R}}^2\right)^m} dr_{\mathrm{R}}. \end{aligned} \tag{60}$$

Let $z = r_{\mathrm{R}}^2$ $(0 \le z \le D^2)$, then $dz = 2r_{\mathrm{R}} dr_{\mathrm{R}}$ and $r_{\mathrm{R}} = \sqrt{z}$. Therefore, using [28, eq. (3.194.1)], we can rewrite last equation as

$$\begin{aligned} T_{in2} &= 2^{m-p-1} \sum_{l=0}^{p} \binom{p}{l} \frac{D^{m+p-2l+2}}{m+p-2l+2} \int_0^{D^2} \frac{z^{\frac{2l+m-p}{2}}}{\left(z + H^2\right)^m} dz \\ &= \sum_{l=0}^{p} \binom{p}{l} \frac{2^{m-p}}{m+p-2l+2} \cdot \frac{1}{2l+m-p+2} \cdot \frac{D^{2m+4}}{H^{2m}} \\ &\quad \cdot {}_2F_1\left(m, \frac{2l+m-p}{2} + 1; \frac{2l+m-p}{2} + 2; -\frac{D^2}{H^2}\right), \end{aligned} \tag{61}$$

where ${}_2F_1$ is Gauss hypergeometric function, as defined by [28, eq. (9.100)].

Then, the proof is completed.

References

[1] P. H. Pathak, X. Feng, P. Hu, and P. Mohapatra, "Visible light communication, networking, and sensing: A survey, potential and challenges," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2047–2077, 4th Quart., 2015.

[2] Z. Ding, K. K. Leung, D. L. Goeckel, and D. Towsley, "On the application of cooperative transmission to secrecy communications," *IEEE J. Sel. Areas Commun.*, vol. 30, no. 2, pp. 359–368, Feb. 2012.

[3] Z. Ding, M. Xu, J. Lu, and F. Liu, "Improving wireless security for bidirectional communication scenarios," *IEEE Trans. Veh. Technol.*, vol. 61, no. 6, pp. 2842–2848, Jul. 2012.

[4] Q. Li, Y. Yang, W.-K. Ma, M. Lin, J. Ge, and J. Lin, "Robust cooperative beamforming and artificial noise design for physical-layer secrecy in AF multi-antenna multi-relay networks," *IEEE Trans. Signal Process.*, vol. 63, no. 1, pp. 206–220, Jan. 2015.

[5] G. Pan, C. Tang, T. Li, and Y. Chen, "Secrecy performance analysis for SIMO simultaneous wireless information and power transfer systems," *IEEE Trans. Commun.*, vol. 63, no. 9, pp. 3423–3433, Sep. 2015.

[6] G. Pan *et al.*, "On secrecy performance of MISO SWIPT systems with TAS and imperfect CSI," *IEEE Trans. Commun.*, vol. 64, no. 9, pp. 3831–3843, Sep. 2016.

[7] H. Shen, W. Xu, and C. Zhao, "QoS constrained optimization for multi-antenna AF relaying with multiple eavesdroppers," *IEEE Signal Process. Lett.*, vol. 22, no. 12, pp. 2224–2228, Dec. 2015.

[8] G. Pan, C. Tang, X. Zhang, T. Li, Y. Weng, and Y. Chen, "Physical-layer security over non-small-scale fading channels," *IEEE Trans. Veh. Technol.*, vol. 65, no. 3, pp. 1326–1339, Mar. 2016.

[9] A. Mostafa and L. Lampe, "Physical-layer security for MISO visible light communication channels," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 9, pp. 1806–1818, Sep. 2015.

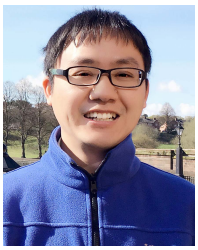
[10] A. Mostafa and L. Lampe, "Optimal and robust beamforming for secure transmission in MISO visible-light communication links," *IEEE Trans. Signal Process.*, vol. 64, no. 24, pp. 6501–6516, Dec. 2016.

[11] A. Mukherjee, "Secret-key agreement for security in multi-emitter visible light communication systems," *IEEE Commun. Lett.*, vol. 20, no. 7, pp. 1361–1364, Jul. 2016.

[12] H. Shen, Y. Deng, W. Xu, and C. Zhao, "Secrecy-oriented transmitter optimization for visible light communication systems," *IEEE Photon. J.*, vol. 8, no. 5, Oct. 2016, Art. no. 7905914.

Authorized licensed use limited to: TU Hamburg-Harburg. Downloaded on November 21,2024 at 11:01:45 UTC from IEEE Xplore. Restrictions apply.

[13] G. Pan, J. Ye, and Z. Ding, "On secure VLC systems with spatially random terminals," *IEEE Commun. Lett.*, vol. 21, no. 3, pp. 492–495, Mar. 2017.
[14] G. Pan, H. Lei, Y. Yuan, and Z. Ding, "Performance analysis and optimization for SWIPT wireless sensor networks," *IEEE Trans. Commun.*, vol. 65, no. 5, pp. 2291–2302, May 2017.
[15] L. Wei, H. Zhang, and B. Yu, "Optimal bit-and-power allocation algorithm for VLC-OFDM system," *Electron. Lett.*, vol. 52, no. 12, pp. 1036–1037, 2016.
[16] F. Wang, Z. Wang, C. Qian, L. Dai, and Z. Yang, "Efficient vertical handover scheme for heterogeneous VLC-RF systems," *IEEE/OSA J. Opt. Commun. Netw.*, vol. 7, no. 12, pp. 1172–1180, Dec. 2015.
[17] S. Shao *et al.*, "Design and analysis of a visible-light-communication enhanced WiFi system," *IEEE/OSA J. Opt. Commun. Netw.*, vol. 7, no. 10, pp. 960–973, Oct. 2015.
[18] Y. Wang and H. Haas, "Dynamic load balancing with handover in hybrid Li-Fi and Wi-Fi networks," *J. Lightw. Technol.*, vol. 33, no. 22, pp. 4671–4682, Nov. 15, 2015.
[19] X. Li, R. Zhang, and L. Hanzo, "Cooperative load balancing in hybrid visible light communications and WiFi," *IEEE Trans. Commun.*, vol. 63, no. 4, pp. 1319–1329, Apr. 2015.
[20] X. Bao, X. Zhu, T. Song, and Y. Ou, "Protocol design and capacity analysis in hybrid network of visible light communication and OFDMA systems," *IEEE Trans. Veh. Technol.*, vol. 63, no. 4, pp. 1770–1778, May 2014.
[21] H. Haas, "Visible light communication," in *Proc. Opt. Fiber Commun. Conf. Exhibit. (OFC)*, Los Angeles, CA, USA, 2015, pp. 1–72.
[22] J. S. Ramos *et al.*, "Towards energy-autonomous wake-up receiver using visible light communication," in *Proc. CCNC*, Las Vegas, NV, USA, 2016, pp. 544–549.
[23] H. Wang, X. Chi, and L. Zhao, "Energy adaptive MAC protocol for IEEE 802.15.7 with energy harvesting," *Optoelectron. Lett.*, vol. 12, no. 5, pp. 370–374, Sep. 2016.
[24] T. Rakia, H.-C. Yang, F. Gebali, and M.-S. Alouini, "Optimal design of dual-hop VLC/RF communication system with energy harvesting," *IEEE Commun. Lett.*, vol. 20, no. 10, pp. 1979–1982, Oct. 2016.
[25] J. M. Kahn and J. R. Barry, "Wireless infrared communications," *Proc. IEEE*, vol. 85, no. 2, pp. 265–298, Feb. 1997.
[26] H.-S. Kim, D.-R. Kim, S.-H. Yang, Y.-H. Son, and S.-K. Han, "An indoor visible light communication positioning system using a RF carrier allocation technique," *J. Lightw. Technol.*, vol. 31, no. 1, pp. 134–144, Jan. 1, 2013.
[27] M. K. Simon and M.-S. Alouini, *Digital Communications Over Fading Channels*, 2nd ed. New York, NY, USA: Wiley, 2005.
[28] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series and Products*, 7th ed. San Diego, CA, USA: Academic, 2007.
[29] E. Damosso, Ed., *Digital Mobile Radio: Towards Future Generation Systems*. Brussels, Belgium: European Commission, 1998, ch. 4.
[30] L. M. Correia, "A view of the COST 231-Bertoni-Ikegami model," in *Proc. 3rd Eur. Conf. Antennas Propag.*, Berlin, Germany, Mar. 2009, pp. 1681–1685.
[31] A. P. Prudnikov, Y. A. Brychkov, and O. I. Marichev, *Integrals and Series: Elementary Functions*, vol. 1, 3rd ed. New York, NY, USA: Gordon and Breach, 1992.
[32] M. Z. Bocus, C. P. Dettmann, and J. P. Coon, "An approximation of the first order Marcum Q-function with application to network connectivity analysis," *IEEE Commun. Lett.*, vol. 17, no. 3, pp. 499–502, Mar. 2013.
[33] S. András, A. Baricz, and Y. Sun, "The generalized Marcum Q-function: An orthogonal polynomial approach," *Acta Univ. Sapientiae Math.*, vol. 3, pp. 60–76, 2011.
[34] S. Andras, A. Baricz, and Y. Sun. (Dec. 22, 2010). The generalized Marcum Q-function: An orthogonal polynomial approach." [Online]. Available: https://arxiv.org/pdf/1010.3348v2.pdf
[35] A. P. Prudnikov, Y. A. Brychkov, and O. I. Marichev, *Integrals and Series: Special Functions*, vol. 2, 3rd ed. New York, NY, USA: Gordon and Breach, 1992.
[36] Z. Ding, K. K. Leung, D. L. Goeckel, and D. Towsley, "Opportunistic relaying for secrecy communications: Cooperative jamming vs. relay chatting," *IEEE Trans. Wireless Commun.*, vol. 10, no. 6, pp. 1725–1729, Jun. 2011.



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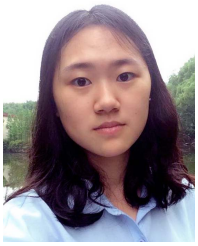

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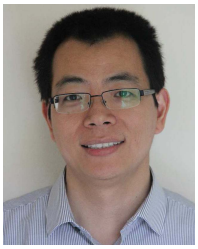

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